

Market Fragmentation

Learning Objectives:

- Market fragmentation
- Costs and benefits of market fragmentation
- How these costs and benefits relate to liquidity externalities
- Implications for regulation of competition among trading platforms

7.1 Introduction

It is common for a security to trade in several venues. For instance, large companies often have their primary listing in their home country and a secondary listing abroad (the Italian energy company ENI, for instance, is listed not only on the Italian Stock Exchange but also on the NYSE). Moreover, several trading platforms permit trading of securities listed elsewhere. For instance, in the United States, stocks listed on Nasdaq or the NYSE can be traded on a myriad of other trading platforms: Cboe's four exchanges, IEX, NYSE Arca, and more. These platforms compete fiercely to attract order flow, and the market share of incumbent exchanges has declined steadily in the United States.

Table 7.1 shows the market shares of the main trading venues for U.S. equities in 2019: NYSE and Nasdaq have market shares of less than 20% each, and electronic communication networks (such as the four exchanges run by Cboe), which operate electronic limit order markets, capture about 16%. Trading is also conducted on regional exchanges and venues known as “dark pools” (crossing networks with minimal transparency requirements), while a relatively large fraction of U.S. trading (27%) is “internalized,” executed directly in house by broker-dealer firms (see Chapter 1 for a description of these trading methods).

The same phenomenon has been observed in the EU since 2007 with the implementation of the Market in Financial Instruments Directive (MiFID), the regulatory framework for European Union securities markets. Prior to MiFID, each member state could require all trading in domestically listed stocks to be executed on the national market. MiFID abolished this “concentration rule,” triggering the entry of new trading platforms (Chi-X, BATS Europe, Turquoise, etc.). Further, the market share of “dark trading”—“dark pools”,

Table 7.1 Market fragmentation in the United States, 2019

Trading Venue	Market share
Nasdaq	17.2%
Nasdaq BX	1.8%
NYSE	13.5%
NYSE Arca	8.4%
Cboe BYX	3.8%
Cboe BZX	5.5%
Cboe EDGA	2.2%
Cboe EDGX	4.8%
IEX	2.7%
Other registered exchanges	2.9%
Broker-Dealer Internalization	27%
Dark Pools (about 33)	10.2%

Source: Securities and Exchange Commission (2020), Staff Report on Algorithmic Trading in U.S. Capital Markets.

over the counter (OTC) trading and systematic internalizers’ (SI) trading—has increased. As a consequence, the market shares of incumbent exchanges (Euronext, LSE, Deutsche Börse, etc.) have declined. The 2021 annual report on EU Securities Markets published by the European Securities Markets Authority (ESMA) estimates that 45% of equity trading occurs outside regulated markets (via OTC trading or SIs), up from 22% in January 2018 and 38% in December 2019. Table 7.2 shows that, as of August 2020, Euronext retained only 66.4% of the trading activity in the CAC40 stocks, and the LSE retained only 61.5% of trading in the FTSE100 stocks.

When a security trades in multiple venues, orders submitted to the different venues do not necessarily contribute jointly to price formation. In this case, the market for the security is said to be “fragmented.” For instance, an aggressively priced sell limit order for AXA on Euronext does not have priority of execution over a simultaneous, higher one in Cboe, a competitor to Euronext for European stock trading. Hence, the Euronext and Cboe markets for AXA operate independently. OTC markets like bond and foreign exchange markets, are generally highly fragmented since they have no mechanisms for consolidating quotes.

Market fragmentation raises several concerns that have been at the forefront of recent regulatory debates, in the United States and in Europe. Market fragmentation may lead to excessive “price dispersion,” meaning that the same security may trade at different prices at the same instant. Worse, outright arbitrage opportunities—the sign of a dysfunctional market—can arise: for instance, the best offer price in one platform may be below the best bid price in another. Such situations of “locked” or “crossed” markets are not at all

Table 7.2 Market fragmentation in Europe, August 2020

Trading Platforms	Market share	
	CAC40 Stocks	FTSE100 Stocks
Euronext	66.4%	n.a.
LSE	n.a.	61.5%
Cboe CXE	16.1%	18.2%
Cboe BXE	4.4%	4.8%
Aquis	6.0%	7.5%
Turquoise	5.0%	7.6%

Source: Cboe Global Markets.

uncommon.¹ In principle, arbitrageurs should exploit these inefficiencies, which should make them both rare and fleeting. If a security trades at two different prices, arbitrageurs can theoretically make a certain profit by buying in one market and reselling in the other. However, there are many reasons why they may decide not to intervene if price differences are small. For instance, for a cross-listed stock, investment or short-sales restrictions may raise the cost of building an arbitrage portfolio. Moreover, arbitrageurs may not monitor all markets continuously and so may react late to transient price divergences. They may even choose to purposefully neglect some arbitrage opportunities, to avoid the risk of being forced to close their arbitrage position prematurely at a loss due to withdrawal of funding or margin calls, as explained in Chapter 11.

Another concern is that market fragmentation could raise trading costs for investors compared to centralized trading for at least three reasons:

- (i) Fragmentation heightens informed investors' ability to exploit their information, allowing them to trade on various platforms and thereby making them harder to detect by other traders (Section 7.2.1). Moreover, by hindering interactions among investors located in different trading venues, fragmentation reduces the scope for risk sharing among investors (Section 7.2.2). Fragmentation also gives local market power to liquidity suppliers in each trading platform, since they do not have to compete with counterparts operating on other platforms (Section 7.2.3).
- (ii) In a fragmented market, investors have to conduct a search for the best price, since quotes are not centralized. This search is costly for several reasons. For instance, it takes time to find the intermediary that posts

¹ For instance, Shkilko, Van Ness, and Van Ness (2008) find that ask and bid prices for NYSE- and Nasdaq-listed stocks that are traded on multiple markets are locked or crossed 10% and 3.5% of the time, respectively. For more recent evidence on this phenomenon, see Wah (2016) and Chapter 9.

the best price. In electronic trading platforms (as in European and U.S. equities markets), the search can be automated by using “smart order routing systems” that scan the platforms for the best price. But these technologies are costly. Search costs also exacerbate agency problems between investors and brokers, as the brokers bear the cost of search while the benefit accrues to their client: fragmentation increases the risk that brokers will not shop around systematically for the best possible price (Section 7.2.4), which may in turn reduce the incentives for liquidity providers to post good quotes (Section 7.4.3).

- (iii) Fragmentation prevents investors from taking full advantage of the “thick market externalities” (also called “liquidity externalities”) that arise from the fact that each additional market participant reduces trading fees or increases liquidity for all other traders. Economies of scale are one source of thick market externalities: in the past, the fixed cost of running a market was high,² so that platforms with large volumes could spread this cost over many traders and so charge lower fees. Thus, concentrating trading in a single market place produced efficiency gains. In today’s securities markets, this argument for concentration is less compelling since automation has considerably shrunk the fixed-cost component of running a market. Another positive externality of market participation is that it makes it more likely to find a trading partner quickly enough: this is easier if the market attracts many traders. Hence, by joining a platform, an investor indirectly increases other market participants’ likelihood of trading and so makes them better off. In Section 7.3, we explore other mechanisms for these externalities, such as reduced adverse selection and improved risk-sharing.

Fragmentation does not only have drawbacks, however. It can benefit investors in various ways. First, competition among platforms tends to reduce fees and foster innovation in trading technology. The recent evolution of US and European equities markets shows the relevance of this argument very well. The entry of new trading platforms (such as BATS in the United States or Chi-X in Europe) drove trading fees sharply down. In the United States, competition has also prompted the NYSE and Nasdaq to overhaul their trading systems. At the turn of the millennium, the NYSE was still predominantly a floor market and Nasdaq was a dealer market; both now have electronic limit order books (LOBs) at their core. Moreover, to serve the trading needs of some clients (in particular, algorithmic traders), trading platforms strive to reduce their “latency” period,

² For instance, traders had to meet in a physical location, which could involve substantial real estate costs.

which is the time it takes to receive and send messages (such as orders or information on standing quotes). For instance, in 2009 the NYSE introduced a new trading system (Super Display Book) that reduced latency from 105 to 5 milliseconds.

Another benefit of intermarket competition is that different trading needs of investors can be best served by different trading mechanisms. For instance, unlike retail investors, institutional investors often execute very large trades, which can have a significant price impact in transparent electronic limit order markets. They accordingly seek to arrange their trades OTC or in dark pools. Section 7.4 explores these benefits of market fragmentation in greater detail.

In the light of these costs and benefits, ideally one would want to organize securities markets so as to promote intermarket competition while allowing interactions between orders placed in different markets. One approach is to let market forces decide on the level of fragmentation. But, it is unlikely that the socially optimal amount of market fragmentation will arise spontaneously. For instance, liquidity externalities act as a barrier to entry, conferring market power on the incumbent markets. Conversely, strategic considerations by trading platforms may lead to excessive fragmentation. For instance, one trading platform might not allow a competitor to access its quotes.³

For this reason, market fragmentation is high on the regulatory agenda. Section 7.5 describes the regulatory framework that rules competition among trading platforms in the United States (RegNMS) and in Europe (MiFID). For instance, the SEC describes the main objectives of Regulation National Market System (NMS), that is, the set of rules under which the U.S. equities market has operated since 2005) as follows:

The NMS is premised on promoting fair competition among individual markets, while at the same time assuring that all of these markets are linked together, through facilities and rules, in a unified system that promotes interaction among the orders of buyers and sellers in a particular NMS stock [...] Accordingly, the Commission [...] has sought to avoid the extremes of: (1) isolated markets that trade an NMS stock without regard to trading in other markets and thereby fragment the competition among buyers and sellers in that stock; and (2) a totally centralized system that loses the benefits of vigorous competition and innovation among individual markets. Achieving this objective and striking the proper balance clearly can be a difficult task. (Regulation NMS, SEC Release No. 34-51808)

³ This situation is similar to that of a firm choosing to make its product compatible or not with its competitors' product. For instance, surcharges can be used to deter depositors who are not clients of a bank from using its ATMs.

One way to reduce fragmentation is to link markets together so that trades for a given security always occur at the best possible price. And this is the approach taken in Regulation NMS. The so-called order protection rule (or trade-through rule) obliges a market center such as an electronic communication network to reroute marketable limit orders to the trading platform that posts the best price when the order is submitted. But this approach has proven very controversial. In particular, it prioritizes execution price over speed of execution, whereas, for some investors, the latter might actually be more important. In Section 7.4.3, we discuss the effects of a trade-through rule on market liquidity.

7.2 The Costs of Fragmentation

In this section, we analyze three channels through which market fragmentation can reduce liquidity compared to centralized trading:

1. asymmetric information (Section 7.2.1),
2. risk bearing capacity (Section 7.2.2),
3. market power (Section 7.2.3).

In all three cases, fragmentation is a source of illiquidity because investors cannot (or do not) access all the possible trading venues for a security: if they could, each platform would simply be a different portal to the same single market. But in practice, multimarket trading is marred by technical hurdles (e.g., differences in time zones or in clearing and settlement systems) and institutional barriers (e.g., divergent tax treatments of groups of investors). Investors may also be unable to split their orders optimally across markets, because this requires costly investment in technology or because their brokers lack the incentive to do so (see Section 7.2.4).

7.2.1 Information Effects

We first analyze the effect of fragmentation in a multimarket version of the model presented in Chapter 4 (Kyle 1985). We assume that the security is traded in two markets, A and B , and that liquidity traders are captive customers of one of the two markets. Let u_A and u_B denote their liquidity demand in markets A and B , and assume that these variables are independently and normally distributed with zero mean and variances σ_A^2 and σ_B^2 . As in Chapter 4, the informed trader knows the final value of the security, v , which is normally distributed with mean μ and variance σ_v^2 . Unlike the liquidity traders, he can trade in both markets, placing an order x_A in market A and x_B in market B . This structure captures the

idea that traders with superior information tend to coincide with sophisticated market professionals (for instance, hedge funds), whose greater trading activity gives them an economic interest in securing access to several markets.

In each market, order imbalances are absorbed by risk-neutral market makers. In market A , market makers set the price p_A conditional on the order flow $u_A + x_A$. The same process occurs in market B . As in Chapter 4, we seek an equilibrium in which market makers post linear price schedules in each market:

$$p_A = \mu + \lambda_A(u_A + x_A), \quad p_B = \mu + \lambda_B(u_B + x_B). \quad (7.1)$$

The optimal order placement strategy for the informed trader (his choice of x_A and x_B) maximizes his expected profits from trading in the two markets:

$$E[x_A(v - p_A) + x_B(v - p_B) \mid v]. \quad (7.2)$$

Substituting into this expression the prices p_A and p_B from equations (7.1), the first-order conditions of the informed trader with respect to x_A and x_B yield the following order placement strategy:

$$x_A = \frac{v - \mu}{2\lambda_A}, \quad x_B = \frac{v - \mu}{2\lambda_B}.$$

Hence, the expression for the informed trader's strategy in each market is the same as in Chapter 4 for a single market. In particular, note that the optimal order size in one market does not depend on the depth of the other. This is because in this setting, an order placed in one market has no impact on the price in the other. Indeed, the informed trader submits orders *simultaneously* in the two markets, so that in market A the price cannot be immediately adjusted in response to concomitant orders in market B , or vice versa.⁴

Proceeding as in Chapter 4, we can then compute the equilibrium values of λ_A and λ_B :

$$\lambda_A = \frac{\sigma_v}{2\sigma_A}, \quad \lambda_B = \frac{\sigma_v}{2\sigma_B}. \quad (7.3)$$

The corresponding equilibrium prices in the two markets are:

$$p_A = \mu + \frac{1}{2}(v - \mu) + \frac{\sigma_v}{2\sigma_A}u_A, \quad p_B = \mu + \frac{1}{2}(v - \mu) + \frac{\sigma_v}{2\sigma_B}u_B. \quad (7.4)$$

⁴ A crucial assumption here is that market makers in one market cannot make their quotes contingent on the order flow in the other, and vice versa. If they could, then the equilibrium would simply be that obtaining in a single market. Hence, the speed of information flows between markets is one determinant of the extent of fragmentation.

Thus, equilibrium prices differ and the two markets are not perfectly integrated. At least in the very short run, such pricing differences are in fact often observed for securities that trade in different markets, even when their trading hours overlap. For instance, Biais and Martinez (2004) document significant price discrepancies between the home and foreign prices for French stocks cross-listed in Frankfurt and for German stocks cross-listed in Paris. Similar evidence is reported for Dutch stocks cross-listed on the NYSE by Hupperts and Menkveld (2002), for Canadian stocks cross-listed in the U.S. by Eun and Sabherwal (2003) and for American Depositary Receipts (ADRs)—claims on non-U.S. stocks traded on U.S. exchanges—by Rösch (2021). Such transient discrepancies suggest that continuous arbitrage is impossible due to the limitations of the trading mechanisms available to market professionals, which prevent them from stepping in to cross offsetting orders in the two markets.

Now consider the depth λ and the price that would prevail in equilibrium if the two markets were consolidated (see Chapter 4):

$$\lambda = \frac{\sigma_v}{2\sqrt{\sigma_A^2 + \sigma_B^2}}, \quad p = \mu + \frac{1}{2}(v - \mu) + \frac{\sigma_v}{2\sqrt{\sigma_A^2 + \sigma_B^2}}(u_A + u_B). \quad (7.5)$$

Thus, the market is deeper when it is not fragmented since $\lambda < \min\{\lambda_A, \lambda_B\}$. The reason is that consolidation increases overall liquidity demand, as measured by $\text{var}(u_A + u_B)$. This effect reduces the informativeness of order flow, and market orders' price impacts are smaller when trading is consolidated in a single market. (This may not hold when liquidity demands in each market are negatively correlated: see exercise 1.)

Thus, fragmentation redistributes the trading gains from liquidity traders to informed traders. In the fragmented environment, the informed trader's profits are on average:

$$E[x_A(v - p_A) + x_B(v - p_B)] = \frac{\sigma_v \sigma_A}{2} + \frac{\sigma_v \sigma_B}{2} = \frac{\sigma_v \sqrt{\sigma_A^2 + \sigma_B^2 + 2\sigma_A \sigma_B}}{2}, \quad (7.6)$$

whereas in a consolidated market they are:

$$E[x(v - p)] = \frac{\sigma_v \sqrt{\sigma_A^2 + \sigma_B^2}}{2}, \quad (7.7)$$

where x is the order placed by the insider and p is the equilibrium price in the consolidated market (see Chapter 4). Comparing these two expressions, it is clear that the informed trader's expected profits are larger in the fragmented market. Since market makers have zero expected profits, the profits of informed

traders are equal to the losses of the liquidity traders. Hence, the increased profits of the informed trader in the fragmented market come at the expense of liquidity traders.

Dealers in one market can better forecast the asset payoff by observing the price set by dealers in the other market. Hence, the price in one market will adjust after a transaction in the other. This process can explain the lead-lag relations that exist between returns of related or identical securities. To see how this works, suppose that after transactions in their respective markets, market makers in one observe the transaction price in the other. Market makers in both markets then adjust their estimate of the value of the security and set their new (long-run) midprice p_l equal to:

$$p_l = E(v|p_A, p_B).$$

Using equation (7.1) and the fact that v , p_A , and p_B are normally distributed, we obtain the long-run price:

$$p_l = \mu + \alpha_A(p_A - \mu) + \alpha_B(p_B - \mu), \quad (7.8)$$

with $\alpha_A = \alpha_B = 2/3$.⁵ This equation shows that market makers' estimate of the fundamental value of the security depends on the informational content of the price changes, $p_j - \mu$, in the two markets. Here the coefficients on these price changes are identical because the informational content of both innovations is identical. In a more general model, the coefficients would differ, with the ratio $\frac{\alpha_j}{\alpha_A + \alpha_B}$ being a natural measure of the "share" of market j in price discovery.

For instance, if α_B is lower than α_A , this means that dealers consider trades (and therefore prices) in market B less informative than those in market A . Hence, market B 's response to a price change in market A is stronger than the reverse. In this sense, market A "leads" market B , or contributes more to price discovery: its "share" of price discovery is higher. Using this intuition, researchers have developed methodologies to estimate the price discovery

⁵ To do this, we must compute the expectation of the security's value v conditional on the two prices, p_A and p_B . Recall that the unconditional expectation is $E(v) = \mu$. Denote by Θ the row vector of covariances ($\text{cov}(v, p_A)$, $\text{cov}(v, p_B)$) and by Ω the variance-covariance matrix of p_A and p_B . From the expressions for the equilibrium prices, we have:

$$\Theta = \frac{\sigma_v^2}{2}(1, 1) \text{ and } \Omega = \sigma_v^2 \begin{bmatrix} 1/2 & 1/4 \\ 1/4 & 1/2 \end{bmatrix}.$$

As v , p_A , and p_B are normally distributed, $E(v|p_A, p_B) = \mu + \Theta\Omega^{-1}(p_A - \mu, p_B - \mu)'$, where the notation $'$ indicates the transpose of a vector. Equation (7.8) follows.

share of a given market for securities that trade on several markets (see Hasbrouck 1995).⁶

One concern about market fragmentation is that it may harm price discovery. This is not the case in the situation posited here. Indeed, consider the variance of the asset payoff conditional on the realization of prices in market A and market B , that is, $\text{var}(v|p_A, p_B)$. This variance measures the uncertainty that still remains for market participants after trading in markets A and B . The lower this variance, the more accurate the price discovery. Calculations show that:⁷

$$\text{var}(v|p_A, p_B) = \frac{\sigma_v^2}{3}. \quad (7.9)$$

Instead, when trading is concentrated in a single market, we have:

$$\text{var}(v|p) = \frac{\sigma_v^2}{2}.$$

Hence, price discovery is better with two markets. The intuition behind this finding is simple: prices in the two markets provide different signals about the value of the security; as these signals are not perfectly correlated (because liquidity trades in each market are not perfectly correlated), they combine to produce a more accurate forecast than the single price signal in a single marketplace.

7.2.2 Risk-Sharing Effects

The previous section examines the cost of market fragmentation consisting in trading costs due to breaking up total liquidity into separate pools. In that model, this effect damages liquidity traders but benefits informed traders to the same extent, since trading is modeled as a zero-sum game. Here, instead, we shall see that the reduction in liquidity caused by fragmentation can generate a net social

⁶ The parameters α_A and α_B can be estimated by a regression of the returns in market A on their own lagged value and the lagged returns in the other market, because equation (7.8) can be rewritten as follows:

$$p_t - p_A = -(1 - \alpha_A)(p_A - \mu) + \alpha_B(p_B - \mu),$$

where the differences $p_t - p_A$ and $p_A - \mu$ are returns in market A over two consecutive periods. Hence, returns in market A are negatively auto-correlated, *conditional* on returns in market B . In contrast, the return $p_t - p_A$ in market A is positively correlated with the return $p_B - \mu$ in market B , because a high return in market B is more likely when the informed investor is buying in this market.

⁷ Let Θ and Ω be as defined in Footnote 5. Then $\text{var}(v|p_A, p_B) = \text{var}(v) - \Theta\Omega^{-1}\Theta'$, which yields equation (7.9).

loss (in the sense that gains may not fully offset losses), because it reduces the extent to which traders can share risks.⁸

As an illustration, consider the model of a competitive call market developed in Chapter 4 (Section 4.3.1), where market makers are assumed to be risk averse, and suppose that the security is traded on two separate call markets. Let the number of market makers be K_A in market A and K_B in market B . For simplicity, we assume that they have no initial inventory ($Z = 0$) and are all equally risk averse, with coefficient $\bar{\rho}$.

Now consider an investor who wants to place a market order of size q on either of the two markets. Using equation (4.25) in Chapter 4 (Section 4.3.1), the equilibrium price at which he will trade in market A is:

$$p_A = \mu + \frac{\bar{\rho}}{K_A} \sigma_v^2 q.$$

Hence, the expected price impact of his order is:

$$\frac{\bar{\rho}}{K_A} \sigma_v^2 |q|,$$

which shows that the liquidity of the market is inversely related to the number of dealers, K_A . Similarly, his expected price impact if he were to trade in market B would be:

$$\frac{\bar{\rho}}{K_B} \sigma_v^2 |q|.$$

If instead the two markets were to merge, the expected price impact on this consolidated trading platform would be:

$$\frac{\bar{\rho}}{K_A + K_B} \sigma_v^2 |q|.$$

Hence, the liquidity of the consolidated market is greater: by increasing the number of market makers, consolidation improves the risk-bearing capacity of the market, and thereby reduces the compensation required by market makers to absorb risky inventories. As in the model of the previous section, consolidation again improves liquidity, but for a different reason.

Market fragmentation not only impairs liquidity but also diminishes the opportunities for risk sharing among market participants. To see this point,

⁸ In Chapter 12, we see that it also increases the cost of capital to firms, and thereby lowers the level of investment in the economy.

consider a situation where the market makers' inventories are unbalanced, so that a reallocation among market makers is Pareto improving. If this reallocation could occur only by trading at competitive prices separately on each of the two markets, then trading would equalize market makers' inventories *within* each market. However, since the participants in the two markets will generally have different per-capita inventory endowments, their holdings after trading would not be equalized *across* the two markets. In that case, there will be further gains from trading if the two markets are merged or fully connected, because there are further opportunities for risk sharing.

The costs of market fragmentation described so far hinge crucially on the assumption that customers cannot access both markets simultaneously, and thus split their orders across them. To illustrate, let us go back to the customer who wants to trade q shares with a market order. Assume now that he can split the order across the two markets by placing an order q_A in market A and an order q_B in market B so that $q_A + q_B = q$. In this case, he splits the order so as to minimize his total trading cost. As the reader can verify by solving exercise 2, the optimal order placement strategy is:

$$q_A^* = \frac{K_A}{K_A + K_B} q, \quad q_B^* = \frac{K_B}{K_A + K_B} q. \quad (7.10)$$

The trader submits a relatively larger order in the deeper market. This strategy equalizes the execution price the two markets since:

$$p_A(q_A^*) = p_B(q_B^*) = \mu + \frac{\bar{\rho}\sigma_v^2}{K_A + K_B} q.$$

Interestingly, the execution price is identical to the price that the customer would obtain if there were a single market. Hence, fragmentation is innocuous if traders can split their orders seamlessly between markets.

These findings suggest two ways of overcoming the costs of fragmentation. The first approach is to merge markets or require concentration of trading in a single market. The second approach consists in facilitating traders' access to all the markets in which a security is traded. Technological advances, such as data aggregators (which consolidate quotes in different markets) and smart order routing technologies (which optimally split orders between trading platforms), lower the cost of multimarket trading. Yet, obstacles remain such as differences in clearing and settlement systems across markets (see Box 7.5), differences in transparency, different trading hours, and agency problems.

Box 7.1 Stock Exchange Mergers and Liquidity

The analysis developed in this section implies that concentration of trading in a single market generates liquidity gains. Some empirical evidence on this point can be garnered from results on the mergers between stock exchanges. Arnold et al. (1999) study the effects of three successive mergers of regional U.S. stock exchanges in the 1940s and 1950s, which transformed them from venues for listing local securities to competitors for the order flow of NYSE-listed companies. The merged exchanges attracted a larger market share and their bid-ask spreads narrowed. A more recent example is the Euronext merger between September 2000 and November 2003: the French, Belgian, Dutch, and Portuguese stock exchanges merged into a single exchange called Euronext, with a single trading platform and a single clearing and settlement system. This consolidation was associated with a significant increase of liquidity, as measured both by trading volume and bid-ask spreads. Following the consolidation, the average spreads on the securities included in the main indices of the Paris, Brussels, Amsterdam, and Lisbon exchanges fell by an estimated 16% to 21%, controlling for other determinants (Padilla and Pagano 2006).

7.2.3 Competition among Liquidity Suppliers

Another cost of market fragmentation is that it hinders competition among liquidity providers, by weakening interactions between market makers active in different liquidity pools and thereby decreasing their incentives to offer good quotes. We illustrate this point below by extending the model analyzed in the previous section to allow for imperfect competition. Then, we analyze a further instance of fragmentation, when market makers can also internalize order execution, that is, execute them against their own inventory at market prices. This effectively turns each market maker into a “separate market.” We show that internalization further reduces competition.

Fragmentation and Market Power

When risk-averse market makers behave as imperfect competitors, the bid-ask spread for investors placing market orders is larger than under perfect competition, as was shown in Section 4.3.2 of Chapter 4. As in the previous section, consider the price impact of a market order of size q on either one of two separate markets A and B , but now assume that each is imperfectly competitive, so that in market A the equilibrium price is:

$$p_A = \mu + \frac{K_A - 1}{K_A - 2} \frac{\bar{\rho}}{K_A} \sigma_v^2 q.$$

Thus, the price impact of an order of size $|q|$ on market A is:

$$\frac{K_A - 1}{K_A - 2} \frac{\bar{\rho}}{K_A} \sigma_v^2 |q|,$$

and similar in market B . If A and B were instead consolidated into a single market with $K_A + K_B$ active market makers, the expected price impact of an order of size $|q|$ would be:

$$\frac{K_A + K_B - 1}{K_A + K_B - 2} \frac{\bar{\rho}}{K_A + K_B} \sigma_v^2 |q|.$$

This expression is lower than its analogue under fragmentation, because the consolidated market has not only greater risk-bearing capacity but also a lower markup. As before, for traders on market A (or market B), the price impact reduction to better risk-sharing is captured by the reduction of the second fraction in the expression from $\bar{\rho}/K_A$ to $\bar{\rho}/(K_A + K_B)$. The further reduction due to more aggressive competition is reflected in the lowering of the first fraction from $(K_A - 1)/(K_A - 2)$ to $(K_A + K_B - 1)/(K_A + K_B - 2)$.

Again, the negative effect of fragmentation on market liquidity would vanish if traders could split their orders between the two markets, or if all market makers were active on both markets simultaneously. In this case, order splitting effectively forces the two pools of market makers to compete against one another.

Internalization and Market Power

Even when trading is centralized, an intermediary can sometimes respond to a client's order in a dual capacity as both broker and dealer. He can choose to channel only a fraction of the order to the market (in his capacity as broker) and take the rest himself (in his capacity as dealer)—a practice known as internalization. In this case, so-called best execution rules require that the broker-dealer executes the internalized order at the current market price or better. This practice is common in the U.S., and has also become increasingly common in Europe, since the MiFID regulation in 2007 allowed it (see Section 1.2.3 and Section 7.5.2). The impact of internalization on market liquidity is controversial.

To analyze this effect, consider the following variant of the previous model. Assume that there is a single centralized market with K imperfectly competitive broker-dealers. One receives a buy market order of size q , and decides what

portion (if any) to internalize and what to execute on the main market.⁹ If he does internalize, he will refrain from selling shares in the main market; otherwise, he would be competing against himself on the internalized portion.¹⁰ Denote by q_I the number of shares internalized and by $q_M = q - q_I$ the remaining shares routed to the market. In this case, the portion of the order routed to the market executes at the following equilibrium price:

$$p(q_M) = \mu + \frac{K-2}{K-3} \frac{\bar{\rho}\sigma_v^2}{K-1} q_M,$$

which is the same expression as in the previous subsection under the assumption that only $K-1$ markets are active.

Now, consider the broker-dealer's optimal strategy, assuming for simplicity that his initial inventory is zero (the case where it is not zero is left as an exercise). The expected utility from routing a buy order for q_M shares to the market is:

$$\begin{aligned} E[U(q_M)] &= E[p(q_M) - \mu] q_I - \frac{\bar{\rho}\sigma_v^2}{2} q_I^2 \\ &= [E(p(q_M)) - \mu] (q - q_M) - \frac{\bar{\rho}\sigma_v^2}{2} (q - q_M)^2. \end{aligned}$$

In increasing q_M , the market maker faces a trade-off, three effects being at play. First, by routing additional shares to the market, he loses the excess return $E(p(q_M)) - \mu$ on each internalized share. Second, by doing so, he increases the price impact for each of the remaining shares and thereby increases the price at which he eventually executes the internalized order (since he will match the market price). Thirdly, by routing more shares to the market, he internalizes less of them and thus decreases his inventory risk. So the first effect of lower internalization reduces his expected utility, while the other two increase it. The best point along this trade-off is found by maximizing the previous expression. The solution to this maximization problem is to route a fraction τ of the order to the market:

⁹ In other words, the internalizer behaves as a Stackelberg leader vis-à-vis all the other broker-dealers, as he chooses the amount executed on the market anticipating the price response by the other broker-dealers.

¹⁰ Selling shares on the main market can never be optimal for him: to see why, suppose that he were to participate to the market alongside with other broker-dealers in filling a portion of the order that he routed to the market. Then, he would put downward pressure on the price and therefore he would lose on the internalized portion of the order. He would, of course, get to sell additional shares in the main market, but he could have sold such shares by internalizing them as well, without affecting the market price adversely.

$$q_M^* = \frac{(K-2) + (K-3)(K-1)}{2(K-2) + (K-3)(K-1)} q \equiv \tau q,$$

and to optimally internalize the remaining fraction $1 - \tau$.

Substituting q_M^* into the expression for $p(q_M)$, we find that, with internalization, the execution price for a market order of size q is:

$$p(q_M^*) = \mu + \frac{K-2}{K-3} \frac{\bar{\rho}\sigma_v^2}{K-1} \tau q.$$

If instead internalization were prohibited, the market maker would have to route the entire order to the market and compete with the other dealers if he wished to be a counterparty to the order. As the previous section shows, the expected execution price for the order in this case would be:

$$p_A = \mu + \frac{K-1}{K-2} \frac{\bar{\rho}\sigma_v^2}{K} q.$$

It is readily confirmed that under internalization the price impact of the order, i.e., $\frac{K-2}{K-3} \frac{\bar{\rho}\sigma_v^2}{K-1} \tau$, is greater than when internalization is banned, in which case the price impact is $\frac{K-1}{K-2} \frac{\bar{\rho}\sigma_v^2}{K}$. For instance, if $K = 4$ and $\bar{\rho}\sigma_v^2 = 10$, the price impact of an order is 4.76 per share when internalization is authorized compared to 3.75 when it is banned.

There are two reasons for this result. First, the dealer who internalizes the order does not take part in filling the portion of the order that he routes to the market. This effectively reduces competition for this order and works to increase the rents earned by all dealers, including himself (on the internalized portion of the order). Second, the dealer strategically executes a significant fraction of the order on the market in order to amplify its market impact, thereby increasing his rent. However, the dealer takes a larger position than he would in the absence of internalization. Indeed, in that case, the dealer would on average absorb a fraction $1/K$ of the order in equilibrium. Under internalization, instead, he absorbs a fraction $1 - \tau$ of the order, which can be shown to be larger than $1/K$. The dealer is willing to take more risk in equilibrium (that is, take a larger position) because internalization allows him to trade at a less competitive price and therefore to obtain a larger expected return per share than without internalization.

Notice that the illiquidity of the main market increases a broker-dealer's ability to extract rents through internalization. Conversely, his scope for rent extraction is minimal if the main market is very liquid: when the number of market makers (K) is very large, τ tends to 1, so that the fraction of the order

internalized tends to zero. In short, internalization impairs market quality, and the damage is greater when the main market is more illiquid. Besides reinforcing dealers' market power, internalization can also impact on liquidity in a context of asymmetric information, if internalizers skim off uninformed order flow (because of a similar mechanism, see exercise 3). For a detailed discussion of this point, see Section 8.4.

7.2.4 Fragmentation and the Broker-Client Relationship

When the market is fragmented, it is harder for investors to locate good prices. The proliferation of “dark pools,” hidden orders on traditional exchanges and crossing networks of unknown depth have compounded the problem.

Thus, in fragmented markets, finding the best trade is more costly and time consuming, a task investors often delegate to their brokers. This delegation creates an agency problem. That is, brokers incur the costs of searching for and implementing the best execution strategy for their clients (which requires technological investment, expertise, and time) without necessarily being able to entirely charge this cost to these clients. As usual in any agency relationship, one should expect brokers to expend less effort in finding the most desirable execution strategy than would be optimal for clients.

Performance-based contracts cannot fully solve this incentive problem for two reasons. First, clients often lack the data and expertise to verify whether their broker did indeed use the best possible strategy to execute their order, especially when the market is fragmented. Second, a trading strategy that is optimal *ex ante* may prove disastrous *ex post*, just because of bad luck. Say, a broker decides to split a client's buy orders into multiple lots to reduce the immediate impact of the order. For large orders, this strategy is often optimal *on average*, but the realized cost in any particular instance is random, as it depends on the evolution of prices over the period during which the order is executed. If prices have risen, the actual average execution price for the client's buy order might prove to be considerably higher than the client could have gotten by executing his order more aggressively. Yet, this poor performance does not indicate a lack of effort or expertise. This example simply shows that it is hard for clients to disentangle what depends on the broker's effort and what is due to bad luck. In this situation, writing incentive contracts is difficult, as one wants to penalize brokers for shirking but not for events beyond their control. To alleviate these agency problems, regulators often require brokers to comply with “best execution rules,” which codify how clients' orders must be handled (see Section 7.5).

Brokers' interests may also conflict with those of their clients because they sometimes receive monetary inducements for routing orders to specific markets or dealers. In particular, in the United States, dealers or exchanges can make payments to brokers who route specific orders to them (e.g., small orders or orders from retail investors).¹¹ A study by the SEC (2000) investigated whether this practice, known as "payment for order flow" (PFOF), affects order-routing decisions in U.S. options markets. Until 1999, there was not much competition among U.S. options exchanges, stock options often being traded only on the exchange where they were listed. For instance, Dell options were listed on the Philadelphia Stock Exchange (PSX) and traded exclusively there. This situation changed in August 1999 when the Chicago Board Options Exchange (CBOE) and the American Stock Exchange (AMEX) announced that they would start trading these options as well. The decision triggered a very fierce battle for market share. As a result, option exchanges started making payments for order flow. The SEC (2000) study shows that by August 2000, 78% of retail customers' orders were routed pursuant to PFOF arrangements. This study further found that PFOF has an impact on routing decisions: the brokerage firms accepting the payments rerouted their customers' orders to the exchanges that paid for order flow (and away from those that did not) much more often than did other brokers (who were much more likely to direct their orders simply to the exchange with the largest market share in the option).

Battalio and Holden (2001) suggest that competition between brokers will force them to pass payments for order flow to their clients, via either lower commissions or improved services. Consistently with this hypothesis, Battalio, Jennings, and Selway (2001) find that the net cost of trading via brokers who sell the order flow to a major Nasdaq dealer is lower than the net cost of trading using brokers who do not sell order flow. This suggests that brokers may pass at least part of their payments along to their clients. But the evidence is mixed: the SEC (2000) reports that in the options markets "few firms are passing along the benefits of payment for options order flow onto their customers in the form of either reduced commissions or rebates." Ernst and Spatt (2022) find that the conflicts of interest associated with PFOF are particularly strong in option markets—indeed stronger than in equity markets.

In any case, payment for order flow could also result in transfers among the different categories of investors. Indeed, several studies suggest that payments for order flow are used to cream-skim orders from uninformed traders (in particular, those from retail investors). Executing these orders at best quotes is

¹¹ This practice is controversial, both in the U.S. and in the E.U. See "EU lawmaker calls for ban on 'payment for order flow'", *Financial Times*, July 27, 2022, and "SEC aims to stem trading practice of payment for order flow," *Financial Times*, June 8, 2022.

profitable since the bid-ask spread contains a compensation for adverse selection costs (see Chapter 3). However, if a dealer or an exchange attracts these orders by paying for them, it increases the likelihood that other dealers or exchanges will receive a larger fraction of informed orders and so will be more exposed to adverse selection. As a result, the bid-ask spreads offered by these other dealers or exchanges will be wider with payment for order flow than without (see Röell 1990, and exercise 3 in this chapter; for empirical evidence, see Bessembinder and Kaufman 1997). If this is the case, the investors whose brokers do not receive payments for order flow will ultimately be worse off, and even those whose brokers do receive payments might be worse off if the practice widens the spread enough. Again, there is conflicting evidence on this issue. For instance, Easley, Kiefer, and O'Hara (1996) find that orders routed pursuant to payments do show larger effective bid-ask spreads. In contrast, Battalio (1997); Battalio, Greene, and Jennings (1997); and Battalio, Greene, and Jennings (1998) find that the introduction of market makers who pay for order flow in equity markets is associated with a narrowing of bid-ask spreads.

7.3 Liquidity Externalities

As the previous sections observe, order splitting between different market segments can neutralize the potential damage done by market fragmentation. Alternatively, trading naturally gravitates to a single marketplace because each trader benefits from the presence of others. Intuitively, as traders expect a market to be more active, they are more inclined to participate, and by patronizing a market, they reinforce other traders' incentive to join. This liquidity-begets-liquidity effect is a force that works naturally to consolidate trading. Here (Section 7.3.1) we illustrate this point using the asymmetric information model set out earlier. Liquidity externalities carry two important implications. First, they imply that securities markets are subject to market tipping, situations in which order flow migrates swiftly from one platform to another (See Box 7.2). Second, they isolate incumbent markets from competition with strong entry barriers. We discuss these implications and illustrate them with empirical evidence in Section 7.3.2.

7.3.1 Liquidity Begets Liquidity

The model developed in Section 7.2.1 can be easily adjusted to capture the idea that liquidity begets liquidity, by assuming that some liquidity traders have discretion in choice of trading venue. Specifically, suppose that M liquidity

traders may choose to participate either in market A or in market B . The liquidity demand of discretionary trader $k \in \{1, \dots, M\}$, denoted by u_k , is normally distributed with mean zero and variance σ_d^2 . The other liquidity traders are locked into one of the two markets. The total liquidity demand of the “captive” liquidity traders in market A , denoted by u_{Ac} , is normally distributed with mean zero and variance σ_{Ac}^2 . We use a similar notation for the total liquidity demand of those based in market B . Last, we assume that all liquidity demands are independently distributed and that $\sigma_{Ac}^2 > \sigma_{Bc}^2$, that is, the mass of captive investors is larger in market A than in market B .

Suppose that M_A liquidity traders with discretion in their trading location choose market A and $M_B = M - M_A$ discretionary liquidity traders choose market B . Then equation (7.3) in Section 7.2.1 yields the depth of market A , and of market B :

$$\lambda_A = \frac{\sigma_v}{2\sqrt{M_A\sigma_d^2 + \sigma_{Ac}^2}}, \quad \lambda_B = \frac{\sigma_v}{2\sqrt{M_B\sigma_d^2 + \sigma_{Bc}^2}}.$$

Suppose that $M_A\sigma_d^2 + \sigma_{Ac}^2 > M_B\sigma_d^2 + \sigma_{Bc}^2$. In this case, $\lambda_A < \lambda_B$: market A is deeper than market B . A discretionary liquidity trader in market B can get lower trading costs by switching to the deeper market A . This relocation makes market A deeper still and so reinforces the incentive for liquidity traders to gravitate to it. Eventually, this “gravitational pull” or liquidity-begets-liquidity effect leads all the discretionary traders to cluster in market A . This situation is an equilibrium, since no trader now has an incentive to move his liquidity demand to market B .

If $M_A\sigma_d^2 + \sigma_{Ac}^2 < M_B\sigma_d^2 + \sigma_{Bc}^2$, the same reasoning implies that trading will concentrate in market B . Such an equilibrium can only arise if $\sigma_{Ac}^2 - \sigma_{Bc}^2 < M\sigma_d^2$.

Last, if $M_A\sigma_d^2 + \sigma_{Ac}^2 = M_B\sigma_d^2 + \sigma_{Bc}^2$, the two markets have exactly the same depth and discretionary liquidity traders are indifferent between them. Again, the condition for obtaining this third equilibrium can be satisfied only if $\sigma_{Ac}^2 - \sigma_{Bc}^2 \leq M\sigma_d^2$.

Thus, when $\sigma_{Ac}^2 - \sigma_{Bc}^2 \leq M\sigma_d^2$, there are three equilibria.¹² In two of them, discretionary liquidity traders cluster in one market, which is therefore the most liquid. In the third equilibrium, they split equally and the two markets have exactly the same liquidity. This equilibrium is fragile, however: defection by even a single trader from one market to the other tips the balance and leads to concentration of trading there—a phenomenon known as market tipping.

The evolution of trading volume for cross-listed securities is a good illustration of the gravitational pull effect. Halling et al. (2008) study the distribution of trading volume between the domestic and the foreign market for cross-listed

¹² When $\sigma_{Ac}^2 > M\sigma_d^2 + \sigma_{Bc}^2$, the sole equilibrium is such that all trading concentrates in market A .

stocks. They observe that foreign trading is active just after the cross-listing date but decreases dramatically over the subsequent six months. In other words, trading activity “flows back” to the home market in the months after the cross-listing, which is consistent with the presence of liquidity externalities that induce the concentration of trading in a single marketplace.

The analysis in this section also shows that one must be careful in interpreting measures of price impacts as measures of the level of informed trading in a market. For instance, when discretionary liquidity traders concentrate in market A , trades have smaller impacts in market A ($\lambda_A < \lambda_B$). Yet, the informed trader trades more aggressively in this market. Thus, in equilibrium, one can simultaneously observe high liquidity (low price impact) and a high level of informed trading activity. This simply reflects the fact that liquidity and informed traders’ aggressiveness are endogenous variables that are jointly determined. Consistent with this logic, Collin-Dufresne and Fos (2015) find empirically that informed traders’ (activist investors) accumulate shares on days in which liquidity is high. Thus, paradoxically, measures of adverse selection may appear low when the level of informed trading is high.

Box 7.2 Tipping: the Case of MATIF in 1998

The evolution of trading on the Matif, a French bond futures market, is a good illustration of how trading volume can shift quickly from one venue to another. Matif was an open outcry market since 1986. In 1998, the market organizers decided to switch to an electronic market. Floor brokers went on strike to oppose the switch; during the strike, trading migrated to a competing electronic trading system (Eurex) in Germany. After the strike, this trading never migrated back to Matif, despite several coordinated attempts by French banks to reactivate it. This episode can be seen as an instance of multiple equilibria in the model described above, where the closure of Matif—say, market B —induced discretionary liquidity traders to opt for Eurex—market A —independently of their expectations regarding other traders. In other words, the mere suspension of trading on market B permanently coordinated traders’ choices on a different equilibrium with all trading concentrated in market A .

7.3.2 Low-Liquidity Traps

The gravitational pull effect is a force that naturally consolidates trading in securities markets. It implies that it is hard for a new trading platform to cut into the market of an incumbent, even if the platform has the potential to be more liquid.

For instance, in the previous model, trading can concentrate in market B even though liquidity traders with discretion would trade more cheaply if they all reconvened in market A . Indeed, if one liquidity trader expects the others to remain in market B , then he is better off staying there. This expectation is self-fulfilling and inefficiently locks traders into market B . All discretionary liquidity traders would benefit from switching to market A if they could do so collectively, but they individually have no incentive to do so. This coordination problem traps them in a low-liquidity equilibrium.

From the point of view of an exchange that wants to compete for the order flow, this translates into a powerful barrier to entry. To overcome it, entrant venues may disseminate information on their trading volumes to show that they attract some trading and make other potential participants more optimistic about their success. This is one reason why the new electronic platforms often publicize their volume statistics so aggressively. Another strategy is to organize meetings among prospective intermediaries (“users”) to facilitate coordination. For instance, in the previous model, suppose that market A manages to attract at least $M_A > (M\sigma_d^2 + \sigma_{Ac}^2 - \sigma_{Bc}^2)/2\sigma_d^2$ discretionary liquidity traders. In this case, the condition $M_A\sigma_d^2 + \sigma_{Ac}^2 > M_B\sigma_d^2 + \sigma_{Bc}^2$ is satisfied and the balance tips in favor of market A . There is a sharp decline of activity in market B , since all the remaining discretionary traders will also switch to market A . The decline thus becomes permanent, as trading is now concentrated in market A in equilibrium.

7.4 The Benefits of Fragmentation

Section 7.2 portrays the dark side of market fragmentation, and Section 7.3 explains how liquidity externalities may lead to the concentration of trading in a single venue. Yet market fragmentation also has a benefit: it enhances “competition for order flow,” by which markets try to attract buy and sell orders. This competition operates at two levels: between platforms and between liquidity providers operating within different platforms.

7.4.1 Curbing the Pricing Power of Exchanges

The concentration of trading on a single trading platform gives this venue monopoly power. Hence, in the absence of competition, a profit-maximizing exchange will set its fees at the monopolistic level, constrained only by the

elasticity of traders' demand for liquidity services.¹³ The evolution of trading fees in the United States and Europe suggest that this concern is realistic. In fact, the entry of new trading platforms in the United States (BATS) and in Europe (Chi-X or BATS Europe) triggered a cut in the fees charged by incumbent markets. As an example, Box 7.3 chronicles the evolution of fees on Euronext in 2003 and 2004 when Deutsche Börse and then the LSE launched trading platforms that allowed brokers to trade Dutch blue chips listed on Euronext Amsterdam.

The escalation of intermarket competition has also led some trading platforms to offer "liquidity rebates" to brokers who submit limit orders (or the reverse). Table 7.3 indicates that in 2020 most U.S. trading platforms paid brokers who submitted limit orders, but charged them a fee for market orders. The table shows that the revenue per share traded for the platforms is a fraction of a cent. For instance, NYSE charges 27.5 cents per round lot (one hundred shares) to "takers" (investors submitting market orders) and rebates 12 cents to "makers" (those submitting limit orders). Thus, it earns 0.155 cents per share traded. On December 18, 2020, 953,501,858 shares of Tape A stocks were traded on NYSE (about 43% of the day's total trading volume in these stocks). Thus, limit order traders collectively received about \$1.14 million in rebates from NYSE, while market order traders paid about \$2.62 million in fees to NYSE.

Table 7.3 Fees per share in cents per round lot for limit orders (make fee) and market orders (take fee)

	Tape A stocks		Tape B stocks		Tape C stocks	
	Make Fee	Take Fee	Make Fee	Take Fee	Make Fee	Take Fee
NYSE	-12	27.5	-12	27.5	-12	27.5
NASDAQ	-20	30	-20	30	-15	30
Cboe EDGX	-17	27	-17	27	-17	27
Cboe EDGA	30	-18	30	-18	30	-18
Cboe BZX	-20	30	-20	30	-20	30
Cboe BYX	19	-5	19	-5	19	-5

Note: Tape A consists of stocks originally listed on the NYSE, while Tape C are stocks listed on NASDAQ. Tape B includes stocks listed on all other exchanges (so-called regional exchanges).

Source: NYSE, NASDAQ, and Cboe company websites, August 2020.

¹³ Fees will be low only if the elasticity of the demand for trading services is high. The evidence on the elasticity of trading volume with respect to transaction costs is mixed: summarizing the available evidence, Schwert and Seguin (1993) report estimates ranging from as little as -0.25 to as much as -1.35.

Liquidity rebates are similar to the payments for order flow discussed in Section 7.2.4. They differ in two important ways, however. First, they go to liquidity suppliers, not demanders. Second, payments for order flow are usually contingent on some characteristics of the clients (e.g., retail orders), while liquidity rebates are not. Hence, their economic role is likely to be different. In both cases, their effect might depend on whether brokers pass their rebates along to their clients, or charge clients for the fees they pay to platforms. Exercise 5 in Chapter 6 considers the effects of make and take fees on the bid-ask spread.

Competition among trading platforms also forces them to install cutting-edge trading technologies. For instance, competition from electronic communication networks in the United States prompted Nasdaq and the NYSE to revamp their trading mechanisms. Another example is the “latency” war among platforms. Some high-frequency traders (e.g., Virtu Financial, Citadel, Optiver, etc.) specialize in automated market making and therefore play a critical role in the provision of liquidity to other market participants (see Chapter 9). To attract them, platforms have updated their trading systems to reduce latency, that is, the lag between an event on the platform (e.g., submission of an order) and the sending of a message about this event to market participants (e.g., “Your order has been executed”). As a consequence, the time interval between orders and trades is now around one millisecond.

Box 7.3 Rivalry among Stock Exchanges

In May 2003, the Deutsche Börse and the LSE separately announced their intention to launch new trading platforms in the Dutch equity markets. At that time, trading in Dutch stocks was mainly concentrated on NSC, a platform operated by Euronext. Euronext reacted to this competitive threat by repeatedly reducing its fees in 2003 and 2004. For instance, in January 2004, Euronext set at €0.3 its order entry fee (in addition to execution fees). Then it halved this fee for limit orders on April 4, 2004, less than two months before the LSE launched its own trading platform for Dutch stocks, EuroSETS, on May 24, 2004. On that day, Euronext suspended the order entry fee on market orders until the end of July and even offered a rebate on total execution costs for these orders. Eventually, on January 31, 2005, Euronext announced that it would stop charging order entry fees on both market and limit orders and also cut its execution fees. EuroSETS’s market share remained a modest at 3.5%, but the mere presence of this platform was sufficient to trigger a significant decrease in trading fees for Dutch stocks.

7.4.2 Sharper Competition among Liquidity Providers

Beside encouraging competition between platforms, market fragmentation can also intensify competition among liquidity providers, and thereby increase consolidated liquidity in comparison to trading concentrated in a single platform. At first glance, this possibility seems to contradict the analysis set out in Section 7.2.2. There, however, it was assumed that liquidity demanders could not split their orders between markets and that liquidity providers could not operate simultaneously on several markets. These assumptions may not adequately describe today's electronic equities markets, where traders often have the technology to provide liquidity where it is most profitable or to demand liquidity where it is cheapest at any instant in time.

In this section we study the effects of market fragmentation when traders can split their orders freely in the context of platforms that use limit order markets, not call markets as in previous sections of this chapter. At the end of Section 7.2.2, we showed that when platforms operate as call markets, fragmentation has no effect in the presence of order splitting. With limit order markets, this is not the case: here, the coexistence of multiple LOBs results in greater consolidated depth, that is, a greater number of shares posted at each price in the market.

To see this point, consider Figure 7.1, which describes an extension of the model of limit order trading considered in Section 6.2. At time 1, investors can submit their limit orders for a security in two limit order markets, the “incumbent” I and the “entrant” E . The payoff v of the security is realized at time 3 and $E(v) = \mu$. The tick size, Δ , is identical in the two markets, and $A_1 = \mu + \Delta$ is the first ask price on the grid above the expected value of the security. For brevity, we focus on the determination of the number of shares, denoted by Y_I and Y_E , offered at price A_1 in each market. Considering other possible prices for limit orders delivers the same insights (see Foucault and Menkveld 2008).

At time 2, a liquidity trader comes to buy or sell shares with equal probabilities. The cumulative probability distribution of the trade size q is denoted by

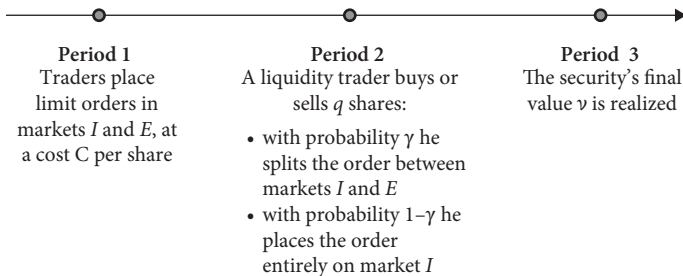


Figure 7.1 Competition between limit order markets: time line.

$F(q)$. For simplicity, we assume that there is no informed trading, but investors submitting limit orders pay an order submission cost C (equal in the two platforms, for simplicity). As in Section 6.2, this cost is the friction that limits the supply of shares at price A_1 .

With probability $1 - \gamma$, the liquidity trader can access market I only. This possibility captures the fact that some brokers may ignore offers posted in some platforms (especially new ones) in order to achieve faster execution or to economize on search costs. Moreover, without an adequate routing technology, monitoring prices in both platforms and splitting orders is cumbersome and time-consuming. With probability γ , the liquidity trader can access both platforms. He then splits his order between the two platforms to minimize the total price impact.

To see how this splitting strategy works, suppose that the liquidity trader wishes to buy q shares of the security. If $q \geq Y_I + Y_E$, he optimally fills all limit orders posted at price A_1 in both trading platforms. The remainder of the order executes against limit order at higher prices. If $q < Y_I + Y_E$, the liquidity trader has several ways to execute the order against the limit orders standing at price A_1 in each market. He may give priority to market I , that is, first buy shares in market I and then consider market E if needed, or else may tap market E first and then buy more shares in market I if necessary. The liquidity trader is indifferent between these routing strategies since they result in the same total payment, $A_1 \times q$. Hence, we assume that the liquidity trader gives priority to market I with probability $\frac{1}{2}$.

As in Section 6.2, the depth at price A_1 in market j (for $j \in I, E$) is determined by a zero-profit condition for the marginal limit order posted at this price in that market. To derive this zero-profit condition, we must first compute the probabilities of execution of the marginal limit order placed at price A_1 in the two markets.

Consider the marginal limit order placed at price A_1 in market I . If the liquidity trader at time 1 only has access to I or gives priority to market I , the marginal limit order at the best ask price in market I executes if and only if $q > Y_I$. Thus, in this case, its execution probability is $1 - F(Y_I)$. If the investor arriving at time 1 can trade in both markets and gives priority to market E , the marginal sell limit order at price A_1 in market I executes if and only if $q > Y_I + Y_E$. In this case, therefore, its execution probability is equal to $1 - F(Y_I + Y_E)$. Hence the unconditional execution probability of the marginal limit order posted at price A_1 in market I is:

$$P_I(Y_I, Y_E; \gamma) = \frac{1}{2} \left[(1 - \gamma + \frac{\gamma}{2})(1 - F(Y_I)) + \frac{\gamma}{2}(1 - F(Y_I + Y_E)) \right]. \quad (7.11)$$

Now consider the marginal limit order posted at price A_1 in market E . If the liquidity trader trades in both markets and gives priority to market E , this marginal limit order executes if and only if $q > Y_E$. If priority is instead given to market I , it executes if and only if $q > Y_I + Y_E$. In all other cases, this marginal limit order does not execute. Hence its unconditional execution probability is:

$$P_E(Y_I, Y_E; \gamma) = \frac{\gamma}{4} [(1 - F(Y_E)) + (1 - F(Y_I + Y_E))]. \quad (7.12)$$

In a competitive equilibrium, the number of shares supplied in each market at price A_1 must be such that no limit order trader finds it profitable to expand the queue of limit orders at this price, in either market. Let $Y_{Ij}^*(\gamma)$ be the number of shares offered in equilibrium at price A_1 in market j . Obviously, if $\gamma = 0$, market E attracts no order since there are no investors routing market orders to it. Thus, $Y_I^*(0)$ is the number of shares offered at price A_1 when all trades take place in market I . As in Section 6.2, if $Y_I^*(0) > 0$, it solves:

$$P_I(Y_I^*(0), 0; 0)(A_1 - \mu) = C,$$

that is, using equation (7.11) for $\gamma = 0$,

$$\frac{1}{2} [1 - F(Y_I^*(0))] = \frac{C}{\Delta}. \quad (7.13)$$

As $1 - F(Y_I^*(0)) < 1$, a necessary condition for the existence of an equilibrium with $Y_I^*(0) > 0$ is $2C < \Delta$. In the rest of this section, we assume that this condition is satisfied, i.e., competitive dealers in market I are willing to sell a positive number of shares at price A_1 if market E is inactive.

Now suppose that $\gamma > 0$. If both markets attract limit orders at price A_1 , then $Y_I^*(\gamma) > 0$ and $Y_E^*(\gamma) > 0$. In this case, the zero-profit conditions on the marginal limit order in the two markets require (again from Section 6.2):

$$P_I(Y_I^*, Y_E^*, \gamma) = \frac{C}{A_1 - \mu} = \frac{C}{\Delta}, \quad (7.14)$$

$$P_E(Y_I^*, Y_E^*, \gamma) = \frac{C}{A_1 - \mu} = \frac{C}{\Delta}, \quad (7.15)$$

For a specific parameterization of the probability $F(\cdot)$, the equilibrium number of shares offered in each market at price A_1 can be computed explicitly by solving the system of equations (7.14) and (7.15) (see exercise 4).

In any case, if the two markets coexist (i.e., both attract limit orders at price A_1), in equilibrium the number of shares offered in each at the best price must be such that their execution probabilities of the marginal limit orders are equal:

$$P_I(Y_I^*, Y_E^*; \gamma) = P_E(Y_I^*, Y_E^*; \gamma).$$

This condition is intuitive; otherwise, the investor who placed the marginal limit order in the market with the smaller execution probability would have an incentive to cancel and resubmit it in the other market at the same price. This would increase his expected profit, since the submission cost is the same.

Figure 7.2 illustrates how the equilibrium is determined, by plotting the number of shares Y_I and Y_E offered in the two markets at price A_1 . The two solid lines in the figure are drawn for $\gamma = 1$, the case in which liquidity traders always access both platforms: the black curve P_I is the set of pairs (Y_I, Y_E) such that the execution probability of the marginal limit order in market I is equal to $\frac{C}{\Delta}$. Similarly, the grey curve P_E is the set of pairs (Y_I, Y_E) such that the execution probability of the marginal limit order in market E is equal to $\frac{C}{\Delta}$. The equilibrium numbers of shares offered at price A_1 in each market are found at the intersection of the two curves. (The two dashed curves instead show how the equilibrium changes when γ decreases below 1, i.e., the case in which liquidity traders cannot always access both platforms, discussed in the next section.)

When $\gamma = 1$, the expressions for the execution probabilities in the two markets are symmetric. Hence, $Y_I^* = Y_E^*$, as shown in Figure 7.2: the number of shares offered at price A_1 is the same in both platforms, so that the total number of shares offered at price A_1 (the consolidated depth at this price) is $2Y_I^*(1)$. Why is there no concentration of trading in a single platform in this case? To answer

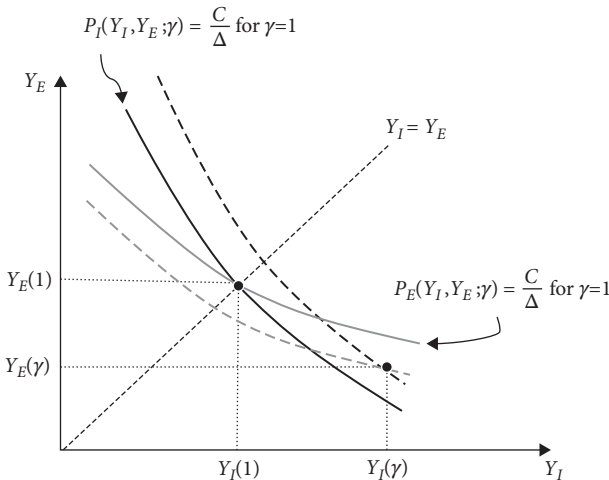


Figure 7.2 Zero-profit depth at the best quote in market I (black) and market E (gray).

this question, suppose that some shares were offered at price A_1 in market I but not yet in market E , and consider an investor who wants to place a limit order at price A_1 . If it is placed in market I , the investor's execution probability is $P_I(Y_I, 0; 1) = \frac{1}{4}(1 - F(Y_I))$. If instead it is placed in market E , the execution probability is $P_E(Y_I, 0; 1) = \frac{1}{4}(2 - F(Y_I)) > \frac{1}{4}(1 - F(Y_I))$. Thus, the investor is better off submitting the limit order at price A_1 in market E , as it yields a higher execution probability: by doing this, the investor jumps the queue of limit orders already posted in market I at the same price, because time priority is not enforced across markets.

Intuitively, this "queue-jumping" possibility intensifies competition among limit order traders, as it allows latecomers to reduce the rents of those who submitted their limit orders first.¹⁴ As a result, the cumulative depth at price A_1 is greater when the two markets coexist, that is, $Y_I^*(1) + Y_E^*(1) > Y_I^*(0)$. To see this, notice that the zero-profit conditions (7.14) and (7.15) imply:

$$\frac{1}{4}[(1 - F(Y^*(1))) + (1 - F(2Y^*(1)))] = \frac{C}{\Delta}, \quad (7.16)$$

where $Y^*(1) \stackrel{\text{def}}{=} Y_I^*(1) = Y_E^*(1)$. Equations (7.16) and (7.13) in turn imply that:

$$F(Y_I^*(0)) = \frac{F(Y^*(1)) + F(2Y^*(1))}{2}. \quad (7.17)$$

As $F(x)$ increases with x , equation (7.17) implies:

$$2Y^*(1) > Y_I^*(0) > Y^*(1).$$

That is, when both markets are active the total number of shares offered at price A_1 (the consolidated depth) is greater, even though the number of shares offered at price A_1 in each market is smaller than when only market I is active. Hence, for traders who can submit market orders in both markets, the coexistence of two limit order markets improves liquidity at the best quotes.

This analysis shows that it can be misleading to consider the effect of market fragmentation on one platform only. For instance, suppose that initially market I operates alone. In this case, the depth of market I at price A_1 is $Y_I^*(0)$. Now suppose that platform E enters and that $\gamma = 1$. We should observe a drop in the liquidity of market I since $Y_I^*(0) > Y^*(1)$. Yet investors submitting market orders smaller than $2Y^*(1)$ and larger than $Y_I^*(0)$ are better off, since they can

¹⁴ The logic is similar to that of the pro-rata allocation rule in a single limit order market: see Section 6.3.2.

execute at a better price. Investors who submit market orders smaller than $Y_I^*(0)$ are indifferent, and those whose market orders are larger than $2Y_I^*(1)$ can be shown to be better off. Indeed, the improvement in liquidity at the best quote is propagated throughout the LOB via the channel just described for the depth at price A_1 . Hence, despite the decrease in liquidity of market I , liquidity demanders are better off because consolidated depth increases at all prices.

The lack of time priority across markets is crucial to this result. The argument also relies on the fact that traders placing limit orders at the head of the queue in each market earn positive expected profits. As explained in Section 6.3.1, this is because the tick size is strictly positive; if it were nil, the number of competing platforms would have no effect on consolidated depth in the framework considered here.

Box 7.4 Competition for Orders and Liquidity: Some Evidence

In 2004, the LSE launched a new limit order market (EuroSETS) to enable Dutch brokers to trade stocks listed on Euronext Amsterdam. Until then, these stocks were traded almost exclusively on NSC, a limit order market operated by Euronext. The launch of EuroSETS accordingly constitutes a good experiment to test whether intermarket competition enhances liquidity in an environment that resembles the one in this section (as NSC and EuroSETS are both limit order markets and their tick size is the same). Foucault and Menkveld (2008) analyze the effects of EuroSETS entry on consolidated depth, using snapshots of the LOBs in each market, and find that consolidated depth increased dramatically, as implied by the model presented in this section.

More evidence on the impact of fragmentation on the liquidity of limit order markets comes from Degryse, de Jong, and Van Kervel (2015), a study of a sample of 52 Dutch stocks (large and mid-cap) in 2006–9. This is an interesting period, since after 2007 the European market became significantly more fragmented owing to the implementation of MiFID (see Section 7.5.2). The sample stocks are listed on Euronext Amsterdam and trade on Chi-X, Deutsche Börse, Turquoise, BATS trading, Nasdaq OMX, and SIX Swiss Exchange. Degryse, de Jong, and Van Kervel (2015) measure market fragmentation using the Herfindahl index, a measure of the dispersion of the trading volume in a stock across the available trading platforms. Their data are very rich, covering limit order books of each trading platform for each stock, so they can study the relationship between their index of market fragmentation and measures of cumulative depth at each price point in all

these limit order books. They find a positive relationship between the market fragmentation and the consolidated liquidity of a stock, as predicted.

7.4.3 Trade-Throughs

Let us compare the case in which liquidity traders do not always have access to both the incumbent and the entrant market ($\gamma < 1$) with the previously analyzed case in which they do ($\gamma = 1$). A decrease in γ reduces the execution probability of the marginal limit order in market E , other things being equal. As a consequence, it becomes less profitable to submit a limit order in market E . But the execution probability of the marginal limit order in market I increases, so it becomes more profitable to submit a limit order there. Thus, in Figure 7.2, the curve that shows the pairs (Y_I, Y_E) such that the execution probability of the marginal limit order in market I is equal to C/Δ shifts upward, while the other curve shifts downward, as shown by the dashed curves. The new equilibrium point is at the intersection of these two curves—clearly, in equilibrium,

$$Y_I^*(\gamma) > Y^*(1) > Y_E^*(\gamma) \text{ when } \gamma < 1.$$

Hence, when $\gamma < 1$ the cumulative depth in market I is greater than when $\gamma = 1$, while that in market E is lower. More generally, the cumulative depth in market E diminishes as γ decreases, and there is a value γ^c such that $Y_E^*(\gamma^c) = 0$. Below this threshold γ^c , no limit order can be profitably posted at price A_1 in market E . Solving for the value of γ such that the solution to the system of equations (7.14) and (7.15) is $Y_E^*(\gamma^c) = 0$ yields $\gamma^c = \frac{4\Delta}{\Delta + 2C}$.¹⁵ It can also be shown that if $\gamma < \min(\gamma^c, 2/3)$, then no limit order will be submitted in this market at any price. Intuitively, in this case the expected revenue from a limit order in market E is too low to cover the cost, because this order is unlikely to execute, due the paucity of investors who track limit orders in market E . Thus, to be active, a trading platform needs some *critical mass* of traders who pay attention to its quotes.

This feature is a barrier to entry for a new platform. To see this, suppose that market E is a new trading platform. It will attract some trades if and only if γ is large enough. Actually, in reality, the value of γ is partly determined by investors' beliefs about the chances of success of the new market. Traders will indeed develop technologies to trade in both markets ("smart routers") if and only if they expect the new market to be liquid enough, so that the reduction in trading costs achieved by accessing both markets exceeds the additional cost

¹⁵ Note that $\gamma^c < 1$, since $2C < \Delta$ by assumption.

of multimarket trading. But, as we have just seen, the liquidity of the entrant market depends critically on the fraction of traders who monitor quotes in both markets. This can lead to a self-fulfilling prophecy, in which traders expect the entrant market to be unsuccessful and so do not consider the offers on it, and—precisely for this reason—market *E* fails to attract liquidity, thus confirming traders' expectations.

In other words, the entry of the new trading platform is blocked by a chicken-and-egg problem: to take off it needs a critical mass of traders following its quotes, but traders will pay attention only if they expect the platform's liquidity to be great enough. For example, this was the key hurdle for Tradepoint, a limit order platform that tried to draw order flow away from the LSE in the late 1990s, as described by Tradepoint's CEO at the time, Nic Stuchfield:

When I was CEO of Tradepoint [...], my team and I spent a considerable amount of effort “selling” the exchange to traders. However, although they all signed up as members, they did not use the market. One major reason was that access to the market was not connected to their trading systems. Even when better bids and offers appeared on our order book, the (momentarily) inferior prices available on the LSE were hit and lifted. Potential users simply could not see, nor easily access the market. If the Tradepoint terminal was at the end of the desk, it was not accessible. The solution [...] was to get Tradepoint integrated into the main order management systems [...] This proved to be easier negotiated than implemented [...] The traders had many other priorities and we could not demonstrate the required liquidity. Think chicken and egg again! (Stuchfield 2003)

This problem can be alleviated if regulation bans violations of price priority between trading platforms—so-called “trade-throughs,” in which a security trades at a worse price in one platform than in another at the same moment. For instance, suppose one hundred shares are offered for sale at $A_1 = \$50$ in markets *I* and *E*. The second best ask price is \$51, at which two hundred shares are offered in each market. Now consider a broker who wants to purchase two hundred shares and plans to execute this trade with a market order. Ignoring the offer in market *E*, the broker buys one hundred shares at \$50 and one hundred shares at \$51 in market *I* only. This routing decision is a “trade-through,” because a buy order executes at \$51 while a sell limit order at \$50 is posted in market *E*.

Trade-throughs do occur in practice. In their study of competition between EuroSETS and NSC (see Box 7.4), Foucault and Menkveld (2008) find that a significant fraction of market orders were executed on NSC (the incumbent) even when EuroSETS (the entrant) offered a better price. Similarly, trade-throughs occur on U.S. equities markets, albeit much less commonly as they are

forbidden (see Hendershott and Jones (2005b)). Indeed, in the United States, the order protection rule (or trade-through rule) require venues to reroute incoming market orders to the market posting the best quote (provided that automatic execution at this quote is possible). This is the order protection rule or trade-through rule, designed specifically to prevent trade-throughs. The rationale is that trade-throughs discourage liquidity provision, as the SEC noted in its June 2005 release of Regulation NMS (Federal Register Vol. 70, No. 124, pp. 37496–644):

Price protection encourages the display of limit orders by increasing the likelihood that they will receive an execution in a timely manner and helping preserve investors' expectations that their orders will be executed when they represent the best displayed quotation. (p. 37505)

This reasoning is consistent with the model developed in this section whereby trade-throughs will occur when $\gamma^c < \gamma < 1$. Consider an investor who wishes to place a market order size equal to $q > Y_I^*(\gamma)$. He should optimally buy part or all of the shares posted at price A_1 in market E . However he will do so only with probability γ . With probability $1 - \gamma$, he will instead buy $q - Y_I^*(\gamma)$ shares at higher prices in market I , a trade-through whose likelihood is therefore $(1 - \gamma)(1 - F(Y_I^*(\gamma)))$. Trade-throughs never happen only if $\gamma = 1$ because in this case all investors optimally execute their market orders against limit orders available in both markets.

Thus, comparing a situation in which $\gamma = 1$ with one in which $\gamma < 1$ is like comparing market structures with and without a trade-through rule. As noted, the number of shares offered in market E is smaller when $\gamma < 1$, because a small γ implies a low likelihood of execution for limit orders placed in market E , and therefore low profitability of limit orders in this market. Hence, a no-trade-through rule effectively encourages investors to submit limit orders in market E by improving their chance of execution.¹⁶

In this model we have analyzed competition between two platforms, taking their design (both being limit order markets) as given. This is of course incomplete. A full-fledged analysis of intermarket competition would have to allow the design of competing trading platforms to be itself endogenous: the venues choose their fees (which could affect C in the previous model) and design their trading mechanisms. In particular, platforms may want to differentiate their mechanisms to relax competition in fees, in the same way that firms differentiate

¹⁶ The model readily generalizes to account for fee differences between trading venues (say, in order submission costs). These fees create asymmetries between the platforms (much like parameter γ does), but do not change the overall logic of the model (see exercise 4).

their products. Such an analysis is needed to see why there is heterogeneity in trading mechanisms and to grasp the logic behind pricing policies such as liquidity rebates.

7.5 Regulation

This chapter begins by setting out the risks and costs of fragmentation: violations of price priority across markets, price dispersion, and diminished liquidity externalities. The second part of the chapter, however, shows that fragmentation also benefits investors by fostering competition among trading platforms. Hence, regulators have the problem of capturing the benefits while lowering the costs of market fragmentation. In this section we explain how the U.S. regulator tried to achieve this difficult balance in designing Regulation NMS and the very different way in which the European Union approached the matter with its MiFID legislation (see also Chapter 1).

7.5.1 Regulation NMS

Regulation NMS went into force in 2006 and 2007.¹⁷ It consists of a series of rules to promote intermarket competition in U.S. equities markets while curtailing the harmful effects of fragmentation.

To understand RegNMS, the historical perspective is helpful. In the 1970s, NYSE-listed stocks traded mainly on the NYSE and not elsewhere. But regional U.S. exchanges and the OTC market gradually captured a larger part of this trading. The U.S. Congress was concerned that this evolution could lead to inefficiencies; in 1975, it mandated the SEC to create a National Market System (NMS). Congress envisioned five purposes for the NMS: (i) economically efficient execution of securities transactions; (ii) fair competition among brokers and dealers and between markets; (iii) availability to brokers, dealers, and investors of information about quotes and transaction; (iv) best execution for investors' orders; and (v) the opportunity to execute orders without the participation of a dealer.

The SEC initially proposed to consolidate all limit orders for each stock in a single file where orders would be executed according to price and time priority. However, this proposal met with strong opposition from exchanges and market makers (see Colby and Sirri 2010). In the end, the SEC opted

¹⁷ It was first released on June 9, 2005. See Securities Exchange Act Release No. 51808, Federal Register Vol. 70, No. 124, pp. 37496–644.

for a more decentralized approach: the NMS would be composed of multiple trading venues, linked together by technology. The resulting market structure that emerged to achieve this goal was based on two pillars: the intermarket trading system (ITS) and the Consolidated Tape Association (CTA). The ITS was intended to make sure that price priority would be enforced across all markets. Hence, an order routed first to one exchange (say the NYSE) would then be rerouted to another (a regional exchange perhaps) if the latter posted a better price, unless brokers or market makers on the first exchange decided to improve the price. Implementation of this no-trade-through rule required real-time information on best bids and offers in every trading venue for a security.¹⁸ The CTA was designed to collect and disseminate this information. Importantly, the no-trade-through rule did not apply to stocks traded on the OTC market, in particular stocks listed on Nasdaq.

However, the developments of technology called for revision of this system. The possibility of computerized trading venues to match buy and sell orders with no need for market makers or floor brokers led to the development of electronic communication networks (ECNs), such as POSIT (a crossing network) and Island (a limit order book) in the 1990s. Speed of execution was a strong selling point of theirs against the more traditional markets, such as Nasdaq and NYSE. The no-trade-through rule, however, required orders to be routed to the NYSE when it was quoting a better price, even though it had slower execution (handled manually by the specialist assigned to each stock). For this reason, several ECNs elected to stay out of the NMS. This became problematic as the ECNs' market share grew, in sync with traders' intensifying demand for rapid execution.

The so-called order protection rule (or trade-through rule) of RegNMS is intended to resolve this problem. The rule extends the protection against trade-through to all NMS stocks, but a quote is protected only if it is immediately and automatically accessible. An implication is that manual quotes sourced by floor-based trading systems (such as the NYSE) were no longer protected against trade-throughs. This new regulatory regime lent further impetus to electronic trading and led to an erosion of the NYSE's market share in its listed stocks. As a result, the NYSE had to overhaul its trading system to ensure its quotes were electronic, so that trade execution would be sufficiently immediate to qualify for order protection.

The rise of the ECNs also prompted platforms to devise new pricing strategies. In particular, ECNs now use the so-called maker-taker pricing model. On a limit order market, each transaction involves a match between a limit order (the "maker") and a market order (the "taker"). Market orders are viewed as

¹⁸ The effects of the no-trade-through rule on market liquidity are analyzed in Section 7.4.3.

consuming (“taking”) the liquidity supplied (“made”) by limit orders. For a platform, the maker-taker model consists in charging a fee for market orders that fill against limit orders on the platform and rebating a fraction of this fee to the filled limit orders (see Table 7.3).

This business model can produce price distortions when combined with the trade-through rule. Liquidity rebates do narrow bid-ask spreads in that they decrease the cost of providing liquidity.¹⁹ Thus, liquidity rebates are a way for a platform to display the best prices in the market more frequently. It does not follow, however, that routing market orders to this platform is optimal because the total trading cost for takers includes the take fee. Yet the trade-through rule applies to quotes, not quotes cum fees. Thus, by granting very generous liquidity rebates, a platform can capture a large share of the order flow (thanks to the trade-through rule) while still earning significant profits by charging a high take fee.

The access rule of RegNMS addresses this problem by capping the take fee (or access fee) at \$0.003 per share. The access rule also prevents platforms from giving preferential treatment (in terms of priority, speed of execution, or fees) to members/subscribers over non-members/subscribers.

Over the years, the CTA’s revenues from the sale of trade and price information became a significant source of income for NMS-members. The allocation of this revenue between members was based on the *number* of trades reported to the consolidated tape, a sharing rule that was creating perverse incentives for platforms. To obtain a larger share, they had an incentive to induce traders to shred their trades into multiple small trades or even engage in wash sales to artificially inflate trading volume.

The market data rules of RegNMS address this issue by changing the revenue-sharing mechanism. The new mechanism is based more on each platform’s contribution to finding the right price. In particular, the new formula rewards the trading venues that frequently set the best bid and ask prices for a given stock.

The last ingredient of RegNMS is the “sub-penny rule,” which imposes a minimum price variation (tick size) of \$0.01 for all NMS securities over \$1.²⁰ The intent was to prevent platforms from competing in the coarseness of their grid size. By setting a very fine grid (with a tick smaller than a penny), a platform could attract limit orders that just barely undercut the best bid or offer price in another market. These orders can generate trades for the platform (due to the trade-through rule), but they gain priority by improving prices for insignificant amounts while undermining the incentives of other traders to quickly post good prices.

¹⁹ They can be viewed as a negative order-processing cost (see Chapter 3).

²⁰ The sub-penny rule does not prevent trading in increments of less than a penny. For instance, traders can agree to match a trade at a price within the best bid and offer price.

RegNMS is designed to secure the benefits of competition without incurring the costs of fragmentation, but it stops short of emulating a nationwide consolidated LOB with multiple points of entry. Indeed, while the regulation ensures that small market orders are executed at the best possible price available on any trading platform, price protection is not given to displayed limit orders that provide market depth at prices outside the best bid and offer. This means that traders who need to fill a large order that cannot be fully executed at the NBBO (after first executing against the protected orders) must hunt around for further market depth, which is often not publicly posted but available in hidden orders on the exchanges or in “dark pools” of liquidity. In this sense, RegNMS does not eliminate all the adverse price disparity and uncertainty effects of fragmentation.

7.5.2 MiFID

Until 2007, E.U. security market regulations allowed member countries to impose “concentration rules”: under the 1993 Investment Services Directive (ISD), member states could require transactions in equity securities to be carried out on a “regulated market.” Therefore some member states—France, Italy, and Spain, among others—maintained rules requiring execution of share trades on their national stock exchange. Others, such as the United Kingdom, left intermediaries free to execute trades off-exchange and also to internalize them, provided they complied with general best execution requirements.

In 2004 the European Union introduced MiFID, whose regulatory regime went into effect on November 1, 2007. The main change relating to fragmentation was a ban on the concentration rules, ushering in free competition between trading platforms. Specifically, MiFID allows three types of trading systems: (i) regulated markets (RMs), (ii) systematic internalizers (SIs), and (iii) multilateral trading facilities (MTFs). Regulated markets are the incumbent exchanges (e.g., NYSE-Euronext and Deutsche Börse). Multilateral Trading Facilities are functionally similar but operate under different regulatory requirements (Cboe or Turquoise are examples of MTFs). Systematic internalizers are “investment firms” (i.e., brokers or banks) that opt to match (“internalize”) buy and sell orders from their clients in-house (either by acting as market makers or by crossing buy and sell orders from different clients).

The abolition of the concentration rules triggered the entry of a series of MTFs.²¹ Meanwhile, anticipating entry by other trading platforms, the existing

²¹ Chi-X (March 2007), Turquoise (March 2007), BATS Europe (April 2008), NasdaqOMX (September 2008), NYSEArca (March 2009), and Burgundy (May 2009).

exchanges pushed for consolidation: the Paris, Amsterdam, Brussels, and Lisbon stock exchanges merged into Euronext; Stockholm's OMX AB acquired stock exchanges in Sweden, Finland, Denmark, Iceland, Estonia, Lithuania, and Latvia; the LSE acquired Borsa Italiana and Turquoise. Many of the new trading platforms and regulated markets also launched “dark pools,” which are simply crossing networks that match buy and sell orders at pre-determined points in time at the midquote set in some other market (see Chapter 1). As of April 2010, the stock and bond trading venues registered in thirty different European countries comprised 127 RMs, 142 MTFs, and 172 SIs.

Thus, with the advent of MiFID, Europe moved closer to the U.S. regulatory framework which encourages competition among platforms. However, there are important differences. With the Order Protection Rule, the U.S. approach interconnects platforms to enforce best execution at the best price nationwide. In Europe, there is no formal order protection rule requiring the routing of market orders to the platform posting the best price. In fact, the notion of cross-market best bid and offer is not yet defined in Europe. Rather, MiFID mitigates the harmful effects of market fragmentation by best execution rules and order handling rules.

More specifically, Article 21 requires that “investment firms take all reasonable steps to obtain, when executing orders, the best possible result for their clients taking into account price, costs, speed, likelihood of execution and settlement, size, nature or any other consideration relevant to the execution of the order,” unless the firm receives a specific instruction from the client. Thus, MiFID does not define best execution only with reference to the price, as RegNMS effectively does with the order protection rule. Consequently, as empirical studies show, trade-throughs happen relatively frequently in European markets.²²

In handling orders, investment firms must abide by transparency rules that mimic the U.S. display rule (Rule 11Ac1-1) and quote rule (11Ac1-4). First, when dealing in shares listed on a regulated market, market makers must immediately disclose any unfilled limit orders so as to make them easily accessible to other market participants, unless instructed otherwise by the customers (MiFID, Article 22). This in effect exposes market makers to competition from public limit orders as in the United States. Second, if they deal in shares traded on regulated and liquid markets, investment firms qualifying as “systematic internalizers” must publish firm quotes on a regular and continuous basis during

²² For instance, Ende and Lutat (2011) estimate the frequency of trade-throughs in the constituent stocks of the Euro Stoxx 50 index traded in eight European markets over twenty trading days in 2007 and 2008 at 12% of the trades in their sample.

normal trading hours (Article 27).²³ These must be firm quotes, in the sense that the intermediary must execute at the quoted prices the orders received from their retail clients: the directive prohibits them from offering price improvements to retail customers and limits the scope of price improvements on large orders from professional clients. This measure is designed to encourage firms to display the liquidity that they offer.

After the implementation of MiFID, European policy makers realized that, in spite these transparency rules, the new regulatory framework had reduced the overall transparency of equity trading in Europe, as the increased competition between platforms induced by MiFID had resulted in “lit markets” (RMs and MTFs) losing market share to more opaque ones (SIs and dark pools). This was a reason for concern, as most of the price discovery process still occurs on lit platforms, with more opaque venues relying on the guidance of prices determined on lit platforms. That opaque platforms free-ride on price formation in lit platforms is witnessed by the fact that their activity typically grinds to a halt when trading stops on their reference lit platforms: for instance, on October 29, 2018, following an unintentional halt of Euronext trading, trading in the French CAC40 stocks almost stopped in most alternative European trading venues, to resume as soon as Euronext trading started again (see Comerton-Forde and Zhong 2021, for systematic evidence regarding the spillover effects of outages across European equity trading platforms).

To address these concerns, in January 2018 the EU enacted the MiFID 2 directive, an updated version of MiFID, and the associated regulation (MiFIR), aimed at moving trading activity back to regulated markets. The updated rules included a mandatory requirement for equity to be traded on RMs, MTFs, or SIs, effectively prohibiting OTC trading of equities. MiFID 2 also mandated a cap on equity trading in dark pools, extended pre-trade and post-trade transparency rules to SIs and investment firms, and broadened their scope to non-equity markets. However, the enactment of MiFID 2 in January 2018 did not result in an increase in the share of trading on lit platforms, as the significant drop in OTC trading triggered by the new rules was largely offset by an expansion of SI.

7.5.3 Comparing the U.S. and E.U. Approaches

To sum up, RegNMS and MiFID are two different regulatory approaches, aimed at the same objective: securing the benefits of intermarket competition for

²³ Transaction reporting to the competent authority can be made by the investment firm itself, a third party representing the investment firm, a reporting system approved by the competent authority, or the MTF through which the investment firm completes the transaction.

investors without the adverse effects of market fragmentation. With its trade-through rule, RegNMS enforces strict price priority. That is, “best execution” is essentially defined with respect to prices. MiFID defines best execution more flexibly since there is no rule against trade-throughs in Europe. In making their routing decisions, brokers must consider price, of course, but can also weigh other dimensions, such as speed and likelihood of execution.

Another important difference between MiFID and RegNMS lies in the consolidation of market data. In the United States, trading platforms are free to disseminate their trade and quote data, but they must also transmit them to an agency that consolidates them across platforms and provides information in real time on the best bids and offers. At the time of writing, such consolidation of market data does not yet exist in Europe, which makes multimarket trading more complicated and costly. A consolidated tape, bundling together data from Europe’s patchwork of trading venues, would in fact require the ability to enforce common technical and operational standards of trade reporting on all trading platforms and punish transgressors—a role that in the U.S. is played by the Financial Industry Regulatory Authority (FINRA).²⁴ So far, in the EU equity market platforms and bond dealers have resisted the introduction of consolidated record collection and provision of standardized data on order flows, transaction sizes, and prices: equity platforms see a consolidated tape as a threat to their revenues from data sales, and bond dealers see it as a threat to the informational rents that they obtain from opacity (see Chapter 8, especially Section 8.5.1). As a result, many fund managers complain that even MiFID 2 rules failed to improve market transparency as intended. This demonstrates that, when security trading occurs simultaneously on various platforms—which by now is typically the case—a sufficient level of transparency is required to overcome market fragmentation.

Yet another difference between the European and U.S. equities markets is that clearing and settlement systems are much less unified in Europe. As a consequence, cross-border trading is more costly for investors than domestic trading (see Box 7.5). This friction hinders multimarket trading (which in Europe is often cross-border trading) and therefore works against the integration of the various trading platforms. For instance, a French institutional investor may find it optimal to trade on Euronext even if a better price is available on Cboe, because the extra clearing and settlement costs of the latter exceed the price gain.

²⁴ FINRA is a U.S. government-authorized, not-for-profit organization that oversees U.S. broker-dealers to provide investor protection, by testing and licensing anyone who sells securities to the public and by ensuring truthfulness of advertising of securities and complete disclosure about the investment product before purchase.

Box 7.5 Clearing Houses and Custodians

Trading platforms enable buyers and sellers to find one another and agree on the terms of a trade. After a transaction, however, each party needs post-trading services.

First, the trade has to be cleared. That is, for each match on the platform, the account of the buyer to which the security needs to be delivered and that of the seller to which the payment is due must be identified. This task is performed by a clearing house or central counterparty (CCP). The CCP also serves to minimize counterparty risk (the risk that one party to the transaction fails to honor its obligation). If one party defaults, the CCP completes the trade, so the other party gets the security or the payment due. For these services, the CCP collects a fee on each trade from the buyer and the seller.

Second, the trade needs to be settled. Settlement—the actual delivery of the security to the seller and its payment by the buyer—usually takes place through a central securities depository (CSD). CSDs also act as custodians for securities, as they keep track of ownership and enable the holders to receive the benefits (dividends, issue rights, etc.).

In the United States, there is a single CCP and CSD for all equity trades. In Europe, there are more than twenty CCPs and CSDs for equities alone, and different platforms use the services of different CCPs or CSDs. Moreover, MTFs have often chosen to connect to incumbent CSDs via new CCPs and agent banks. For instance, Cboe Europe use EuroCCP as its CCP while Euronext uses Euronext clearing. As a consequence, cross-border transactions are more complex and costlier than domestic transactions since they involve more layers of intermediation.

7.6 Further Reading

Several authors have analyzed thick market externalities in securities markets. Mendelson (1982, 1985, 1987) and Hendershott and Mendelson (2000) show how an increase in the number of participants on a platform (a call market in Mendelson (1982, 1985), a crossing network in Hendershott and Mendelson (2000)) raises the likelihood of finding a match for all traders. Pagano (1989b) considers a different source of thick market externality. In his model, risk-averse traders perceive their demand for the stock as adversely affecting the market price. A greater number of market participants implies a lower price sensitivity to each trader's net demand, thus increasing the market's liquidity. The analysis of risk-sharing effects and market fragmentation in Section 7.2.2 is based on Pagano (1989b).

The tangle market also emerges in models with asymmetric information, such as Admati and Pfleiderer (1988) and Chowdhry and Nanda (1991). Admati and Pfleiderer were the first to consider “discretionary liquidity traders,” in a setting in which trading is fragmented across different times of day. Chowdhry and Nanda use a similar model to analyze the fragmentation of trading across different venues; the setting analyzed in Sections 7.2.1 and 7.3.1 is a simplified version of their model.

Parlour and Seppi (2003) develop a model of competition between a dealer market and a LOB. Foucault and Menkveld (2008) extend this model to analyze competition between two limit order markets. The model developed in Section 7.4.2 is based on Foucault and Menkveld (2008). Foucault and Parlour (2004), Colliard and Foucault (2012), Pagnotta and Phillipon (2018), Chao, Chen, and Ye (2019), and Baldauf and Mollner (2021) analyze competition between trading platforms and endogenize their fees.

There is also a rich empirical literature on the effects of market fragmentation and intermarket competition on market liquidity and price discovery. Among others see Foerster and Karolyi (1998), Mayhew (2002); Barclay, Hendershott, and McCormick (2003); Bessembinder (2003); Boehmer and Boehmer (2003); De Fontnouvelle et al. (2003); Battalio, Hatch, and Jennings (2004); Jennings, Boehmer, and Wei (2007); Foucault and Menkveld (2008); Biais, Bisière, and Spatt (2010); O’Hara and Ye (2011); Degryse, de Jong, and Van Kervel (2015); Gresse (2017); Haslag and Ringgenberg (2021).

7.7 Exercises

1. **Market consolidation with correlated noise trading.** Consider the model developed in Section 7.2.1, but assume that u_A and u_B are correlated. Denote by ρ the correlation between these two variables.
 - a. How does market depth depend on ρ when order flow is consolidated in a single market?
 - b. Does order flow consolidation increase or decrease market depth?
 - c. How does the informed investor’s expected profit differ from those obtained in the model of Section 7.2.1?
2. **Optimal order splitting.** Consider an order of total size q that can be split into two orders q_A and q_B to be executed in markets A and B . The security traded in these two markets has expected value μ and variance σ_v^2 . Markets A and B are populated by K_A and K_B competitive dealers with no initial inventories and identical risk aversion $\bar{\rho}$ as in Section 7.2.2. Show that equation (7.10) is the optimal split of the total order q between the two markets.

- 3. Payments for order flow.** Consider the market for a risky security. Its payoff at time 1 is either $v^H = \mu + \sigma$, or, $v^L = \mu - \sigma$ with equal probabilities. At time 0, an investor gives to his broker an order to buy or sell one share of the security to a broker. With probability ϕ , the investor is a retail investor and has no information on the payoff. In this case he buys or sells the security with equal probability. With probability $1 - \phi$, the investor is an institutional investor. In this case, he is perfectly informed about the payoff with probability α , or uninformed with probability $1 - \alpha$. In the latter case, the investor is a buyer or seller with equal probability. Bid and ask quotes for the broker's order are posted by three risk-neutral dealers 1, 2, and 3 before the broker contacts them. The broker cannot split his order among dealers. Dealers have no private information on the payoff of the security. For this exercise, you need also to refer to the material in Section 3.3.2.
- Assume that there is no payment for order flow between the broker and the three dealers. In this case, the broker randomly selects one dealer among those posting the best price for his order. Compute the bid and ask quotes posted by the dealers.
 - Assume now that dealer 1 has a payment for order flow arrangement under which the broker gives dealer 1 all orders from retail investors and the dealer commits to execute all these orders at the best quotes (i.e., the ask and bid price set by the remaining dealers, 2 and 3). Other orders are sent to dealer 2 or 3 (as in question a). What are the quotes posted by dealers 2 and 3? Deduce that the bid-ask spread is higher in this case than where there is no payment for the order flow.
 - Let P be the payment of dealer 1 to the broker. What is the largest possible value of P ?
 - Is payment for order flow beneficial or detrimental to investors?
- 4. Competition between limit order markets with uniformly distributed market orders.** Consider the model of Section 7.4.2 and assume that the size of the market order $\tilde{X}$ has a uniform distribution on $[0, \bar{X}]$. That is, $F(x) = x/\bar{X}$. We denote by $Y_{jk}^*(\gamma)$ the cumulative depth posted at the ask price $A_k = \mu + k\Delta$ in market $j \in \{I, E\}$ when the fraction of investors submitting market orders in both markets I and E is γ , and by c_j be the submission cost in market j .
- Assume that $2c_I \leq \Delta$ and that $\gamma = 0$. Show that the equilibrium cumulative depth at price A_k is:

$$Y_{I1}^*(0) = \bar{X}(1 - \frac{2c_I}{\Delta}). \quad (7.18)$$

- b.** Now suppose that $\frac{4c_E}{\Delta+2c_I} < \gamma$ and that the other parameters are such that $Y_{I1}^*(\gamma) > 0$, $Y_{E1}^*(\gamma) > 0$, but $Y_{I1}^*(\gamma) + Y_{E1}^*(\gamma) < \bar{X}$. Compute $Y_{I1}^*(\gamma)$ and $Y_{E1}^*(\gamma)$ as a function of γ . Deduce further from the result that the conditions $Y_{I1}^*(\gamma) > 0$ and $Y_{E1}^*(\gamma) > 0$ are satisfied if and only if $\frac{4c_I}{\Delta(2-\gamma)+2c_E} < 1$. Moreover deduce that the condition $Y_{I1}^*(\gamma) + Y_{E1}^*(\gamma) < \bar{X}$ is satisfied if $4(\gamma c_I + (2-\gamma)c_E) > (2-\gamma)\gamma\Delta$.
- c.** Deduce from question **b** that the two markets can coexist even if their order submission costs differ and $\gamma = 1$.
- d.** Why does the cumulative depth at price A_1 in one market decrease with the order submission cost in this market but increase with the cost in the competing market?
- e.** Consider the case $\gamma = 1$ and suppose that $4(c_I + c_E) < \Delta$ and $4c_I < \Delta$. Compute $Y_{I1}^*(1)$ and $Y_{E1}^*(1)$.
- f.** Under the assumptions in question (e), what is the number of shares offered at price $A_k > A_1$? Is the result different when $\gamma = 0$?